E1 · 10 HELD-OUT ETH3D SCENES · FROZEN AT 71582C1 (TAG E1-PREREG)

MapAnything Under Wrong Hints

We gave MapAnything wrong hints of 19 kinds (wrong camera positions, wrong lens settings, wrong depth, a forgotten metric flag) and measured whether it ends up worse than with no hints at all.

DECISION (RULES FROZEN BEFORE THE RUN)

Strong GO

Signal	Question	Answer
S1 Harm	Wrong hints make it worse than no hints	YES
S2 Spread	One bad hint damages the good views	YES
S3 Blind trust	The model copies bad hints	YES
S4 Silent	The model's confidence gives no warning	NO
S5 Simple fix fails	The easy filter is not enough	YES
S6 Agrees-but-wrong	Correlated / lens / depth errors hurt, or scale goes wrong unnoticed	YES
S7 Realistic	The harm happens at realistic error sizes	YES

2350 runs on 10 test scenes, 0 failures. 13 of 19 kinds of wrong hint made results worse than no hint at a realistic size. Median "absorption" of a single bad camera: 0.96 (1 = fully copied). Model confidence AUROC for spotting the bad view: 0.91 (0.5 = no signal, 1 = perfect).

What this means, in plain words

> **1. Wrong camera *directions* are the danger.** One camera turned 15° wrong made camera-rotation error 114% worse than giving no poses at all, in all 10 scenes (30°: 321% worse). Every camera jittered by 2°: 52% worse (8/10 scenes). 20% of the cameras wrong: 295% worse (10/10). The first (reference) camera turned 15°: 3D shape 164% worse, rotation 124% worse.

> **2. Errors that agree with each other hurt, and the simple filter cannot see them.** Odometry-style drift (level 2) made rotation 142% worse than no poses (10/10 scenes); the same 5° heading error on every camera, 108% worse (9/10). The "compare with photos-only" filter recovered about 10% of that loss at most. It fixes big single outliers (15° or more: 97-99% recovered), but not a 5° outlier (4%) and not drift or a shared heading error.

> **3. Wrong camera *positions* are mostly harmless.** Position jitter and small glitches still left results 47-83% better than photos only. Only a far jump hurts the 3D shape: 3 scene-lengths, 62% worse in 8 of 10 scenes, but the 95% interval still touches zero, so it does not count as harm; 10 scene-lengths, 348% worse.

> **4. The model copies wrong scale.** Telling it every distance is 1.5× too long made its scene 1.38× too big (95% CI 1.32-1.43); the real-size 3D error rose 75% above photos only (8/10 scenes). Depth hints 1.5× too deep on 2 of 10 views made the 3D shape 139% worse than without depth hints (10/10); one such view alone, 78% worse. Slightly noisy depth (5%) did no harm.

> **5. Wrong lens settings hurt only a little.** A focal length 10% too long on every photo made rotation 18% worse than no lens info (+0.05°, 8/10 scenes), and the run was still 66% better than dropping the camera poses. A forgotten metric flag was not harmful compared with photos only, though the scene came out 16% too big.

> **6. The model copies a bad hint almost completely, even though its confidence drops.** For a single bad camera it took on 96% of the wrong pose (median "absorption" 0.96). Yet its own confidence ranks the bad view lower than the good ones 91% of the time (AUROC 0.91). It has the warning signal but does not use it, which matters for E3.

> **7. Damage spreads to good views only modestly.** One bad camera raised the good views' rotation error by 3-8%; a far jump raised their 3D error by 13%; with 20% of cameras wrong, the good views' rotation error rose 20%, just over the preregistered bar.

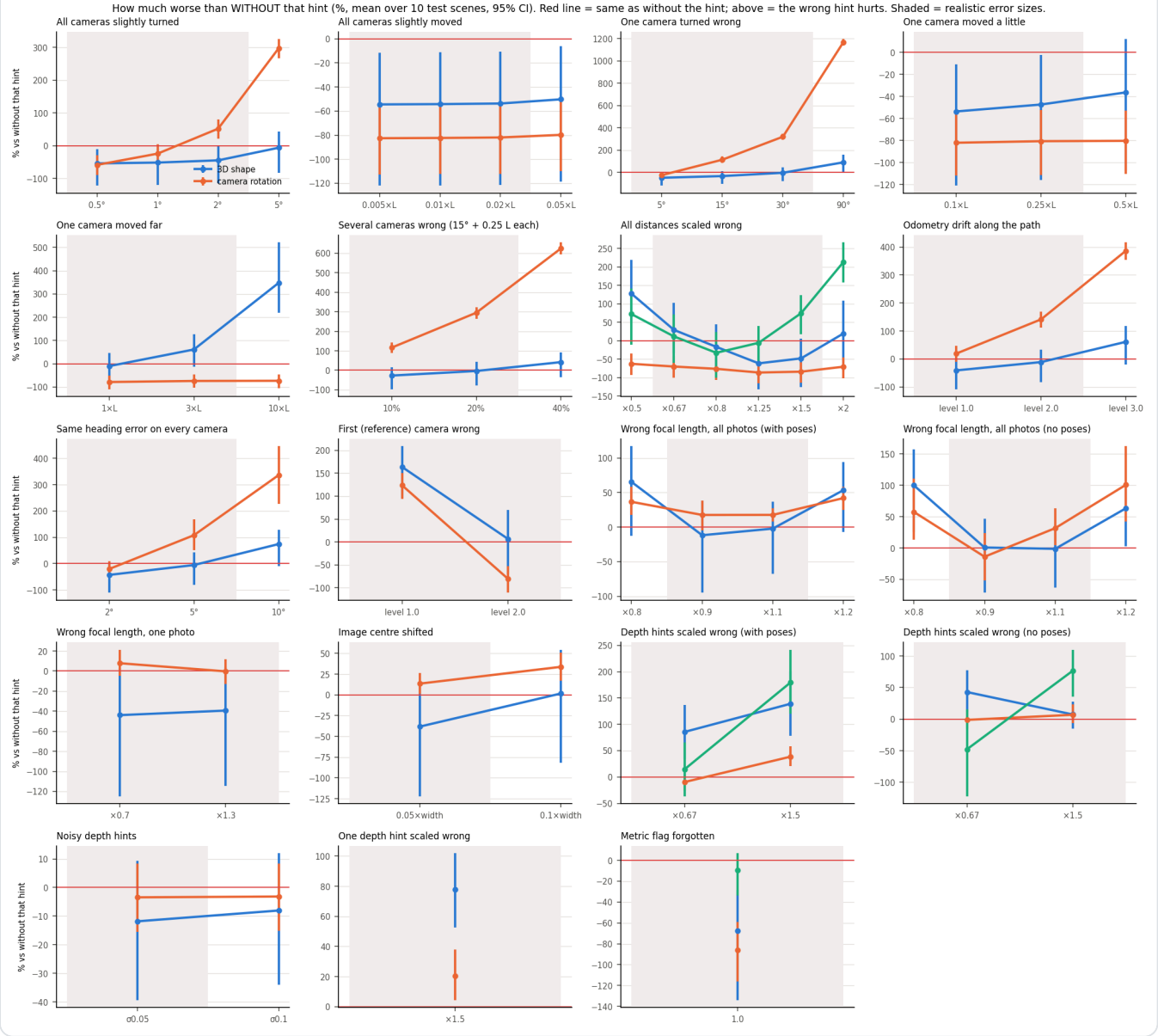
Sources: family_summary.csv, dose_response.csv, fix.csv, contamination.csv, absorption.csv, confidence_auroc.csv, scale.csv (reports/e1_results/). "Worse than" always compares the same scenes with and without the hint.

What we tested

Each test is one scene, 10 photos, rebuilt by MapAnything with hints. Every wrong-hint run is compared with the same scene's runs with **no** hint of that kind (the "floor") and with the **correct** hint (the "ceiling"). Scenes and their size L: courtyard 1.6 m, electro 2.4 m, kicker 1.7 m, meadow 6.8 m, office 0.2 m, pipes 0.9 m, playground 0.7 m, relief_2 1.2 m, relief 1.2 m, terrains 0.8 m.

> **How to read "benefit lost".** 0 = as good as the correct hint, 1 = as bad as having no hint. Above 1 means the wrong hint is worse than no hint. It is a ratio, so when the correct hint barely helps it can get very large from a small change; the tables also show the plain difference.

Headline: how much worse than without that hint



One panel per kind of wrong hint. Blue = 3D shape, orange = camera rotation, green = 3D at real size. Red line = same as without that hint; above it the wrong hint hurts. Shaded columns = realistic error sizes. Mean over the 10 test scenes with 95% confidence intervals.

Each kind of wrong hint, at its worst realistic size

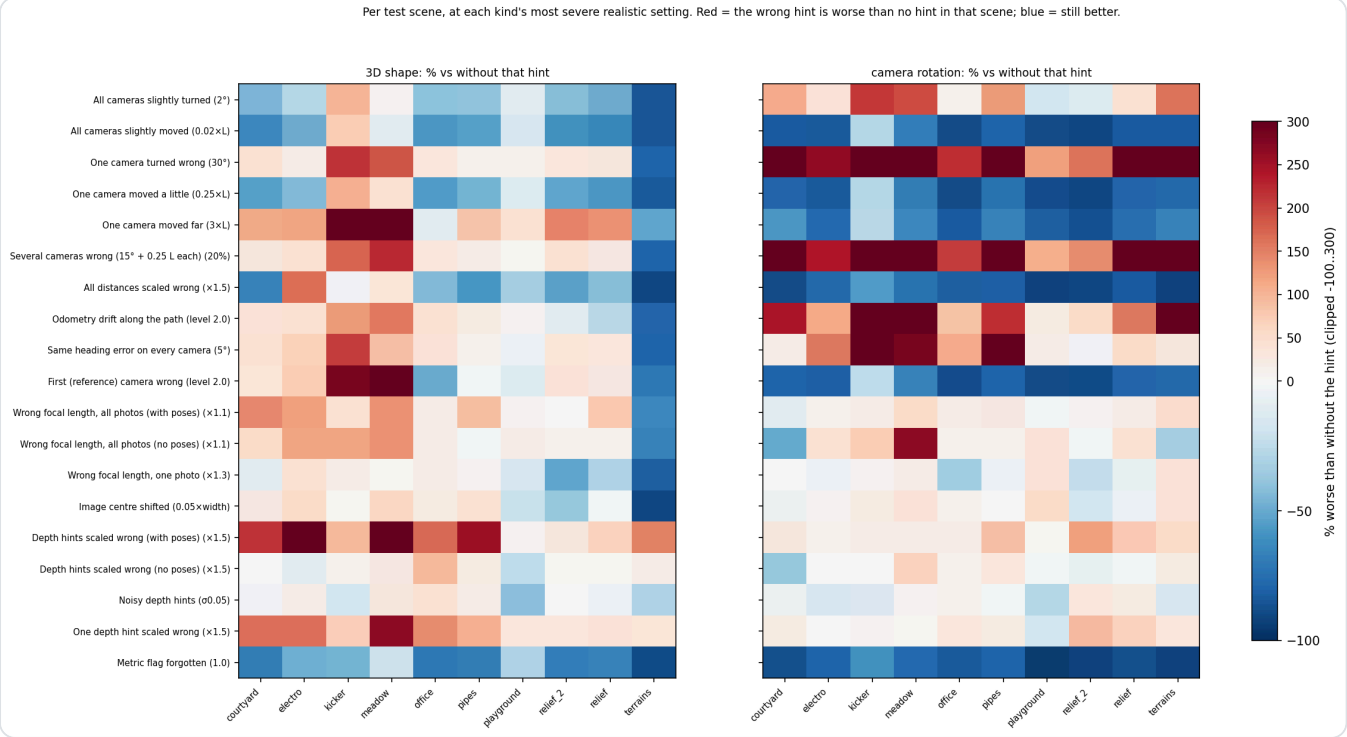
Wrong hint	Worse than no hint?	Setting	3D shape vs no-hint	Rotation vs no-hint	Rotation Δ	Scenes worse	vs dropping poses
All cameras slightly turned jittery tracking	YES	2°	-44.7%	+51.9%	+0.734°	80%	+134.9%
All cameras slightly moved jittery tracking	NO	0.02×L	-53.7%	-81.9%	-1.158°	10%	+15.1%

Wrong hint	Worse than no hint?	Setting	3D shape vs no-hint	Rotation vs no-hint	Rotation Δ	Scenes worse	vs dropping poses
One camera turned wrong bad compass reading / flipped photo	YES	30°	-3.1%	+320.5%	+4.531°	100%	+550.2%
One camera moved a little small GPS glitch	NO	0.25×L	-47.3%	-80.6%	-1.139°	20%	+31.0%
One camera moved far GPS jump	only extreme	3×L	+61.6%	-73.6%	-1.041°	80%	+301.6%
Several cameras wrong (15° + 0.25 L each) bad stretch of sensor data	YES	20%	-3.7%	+295.2%	+4.173°	100%	+511.0%
All distances scaled wrong wrong units / scale drift	YES	×0.67	+29.5%	-69.7%	-0.986°	70%	+221.8%
Odometry drift along the path visual-inertial drift	YES	level 2.0	-11.1%	+141.5%	+2.000°	100%	+273.4%
Same heading error on every camera compass / gravity misalignment	YES	5°	-5.6%	+107.9%	+1.525°	90%	+221.4%
First (reference) camera wrong bad first frame	YES	level 1.0	+163.6%	+123.5%	+1.746°	100%	+555.1%
Wrong focal length, all photos (with poses) wrong EXIF / calibration	YES	×1.1	-2.2%	+17.6%	+0.046°	90%	-66.4%
Wrong focal length, all photos (no poses) wrong EXIF / calibration	YES	×1.1	-1.4%	+31.6%	+0.447°	80%	–
Wrong focal length, one photo one photo with wrong metadata	NO	×0.7	-44.0%	+7.8%	+0.020°	70%	-69.2%
Image centre shifted crop / resize metadata bug	YES	0.05×width	-38.2%	+13.5%	+0.035°	70%	-67.6%

Wrong hint	Worse than no hint?	Setting	3D shape vs no-hint	Rotation vs no-hint	Rotation Δ	Scenes worse	vs dropping poses
Depth hints scaled wrong (with poses) depth sensor / mono-depth scale	YES	×1.5	+139.0%	+38.5%	+0.095°	100%	+170.5%
Depth hints scaled wrong (no poses) depth sensor / mono-depth scale	YES	×0.67	+43.0%	-1.2%	-0.011°	90%	–
Noisy depth hints noisy depth sensor	NO	σ0.05	-11.8%	-3.4%	-0.008°	50%	-74.0%
One depth hint scaled wrong one bad depth map	YES	×1.5	+77.8%	+20.4%	+0.050°	100%	+101.3%
Metric flag forgotten user forgets the flag	NO	1.0	-67.4%	-85.8%	-1.213°	0%	-18.9%

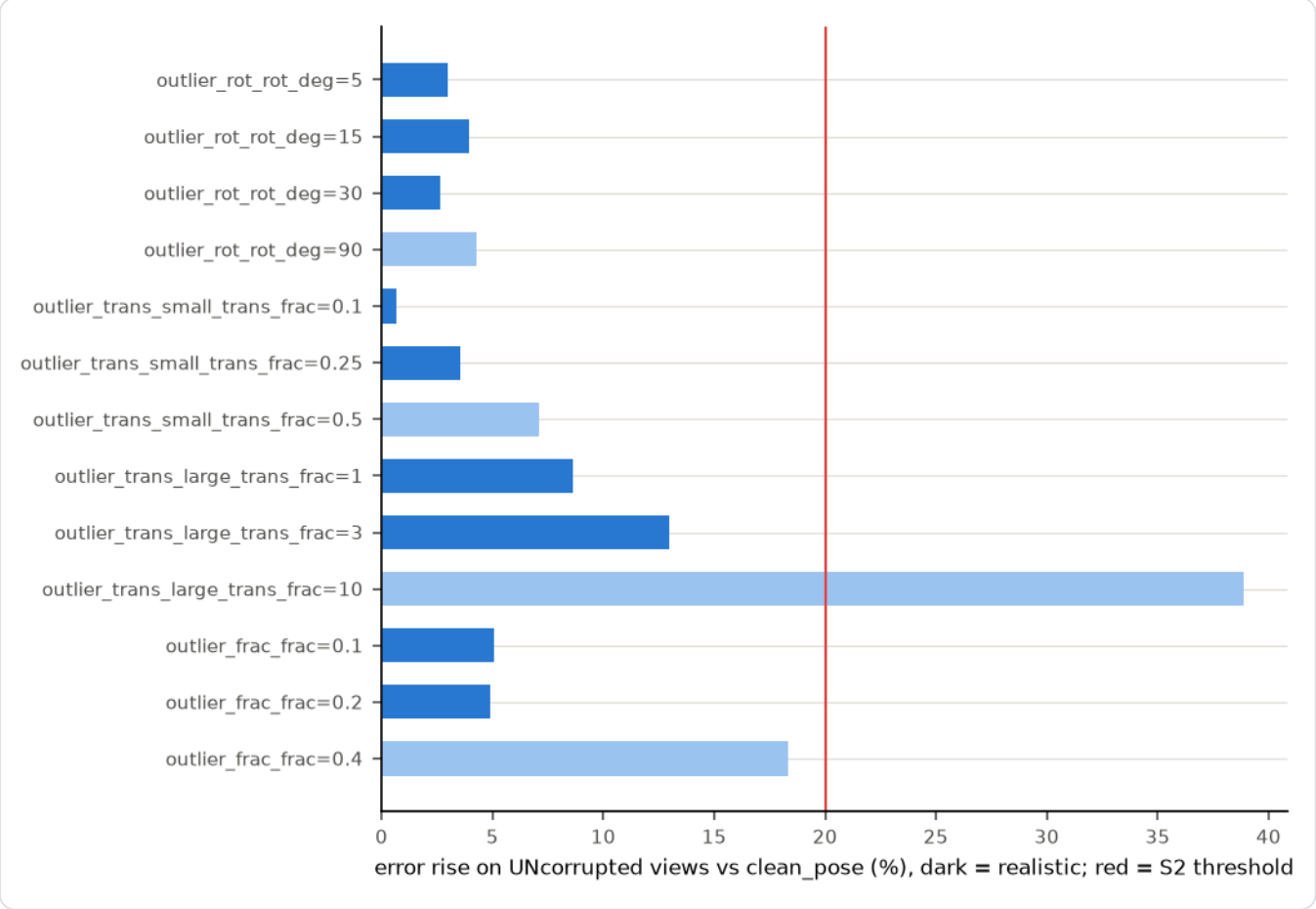
"vs no-hint" = how much worse than the matching floor (+ = worse). "vs dropping poses" = the same run against photos + lens with all poses dropped (+ = dropping the poses would have been better). Source: family_summary.csv, dose_response.csv.

Is it one unlucky scene?



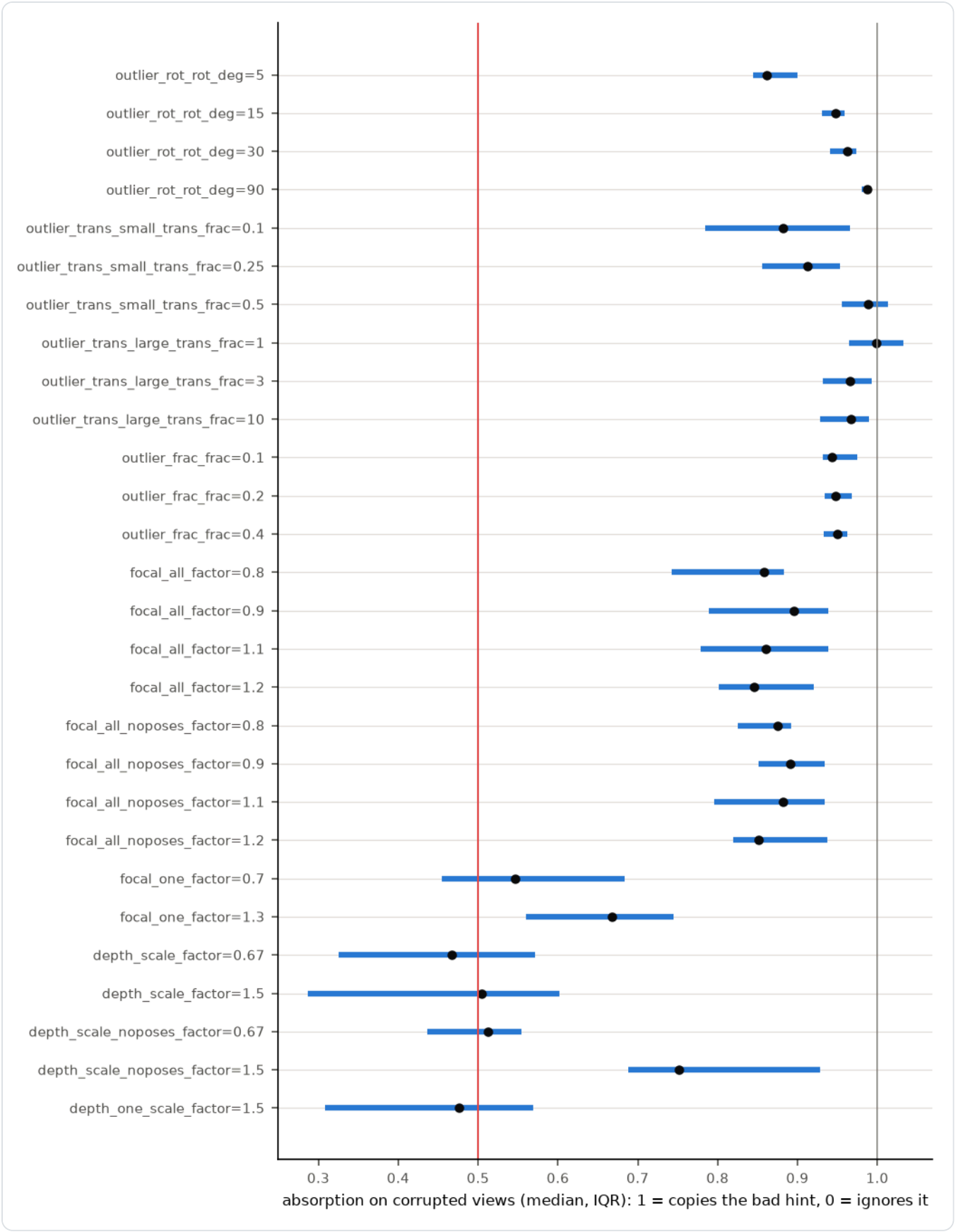
Each cell is one test scene at each kind's most severe realistic setting. Red = in that scene the wrong hint was worse than no hint; blue = still better.

Does one bad hint hurt the good views?



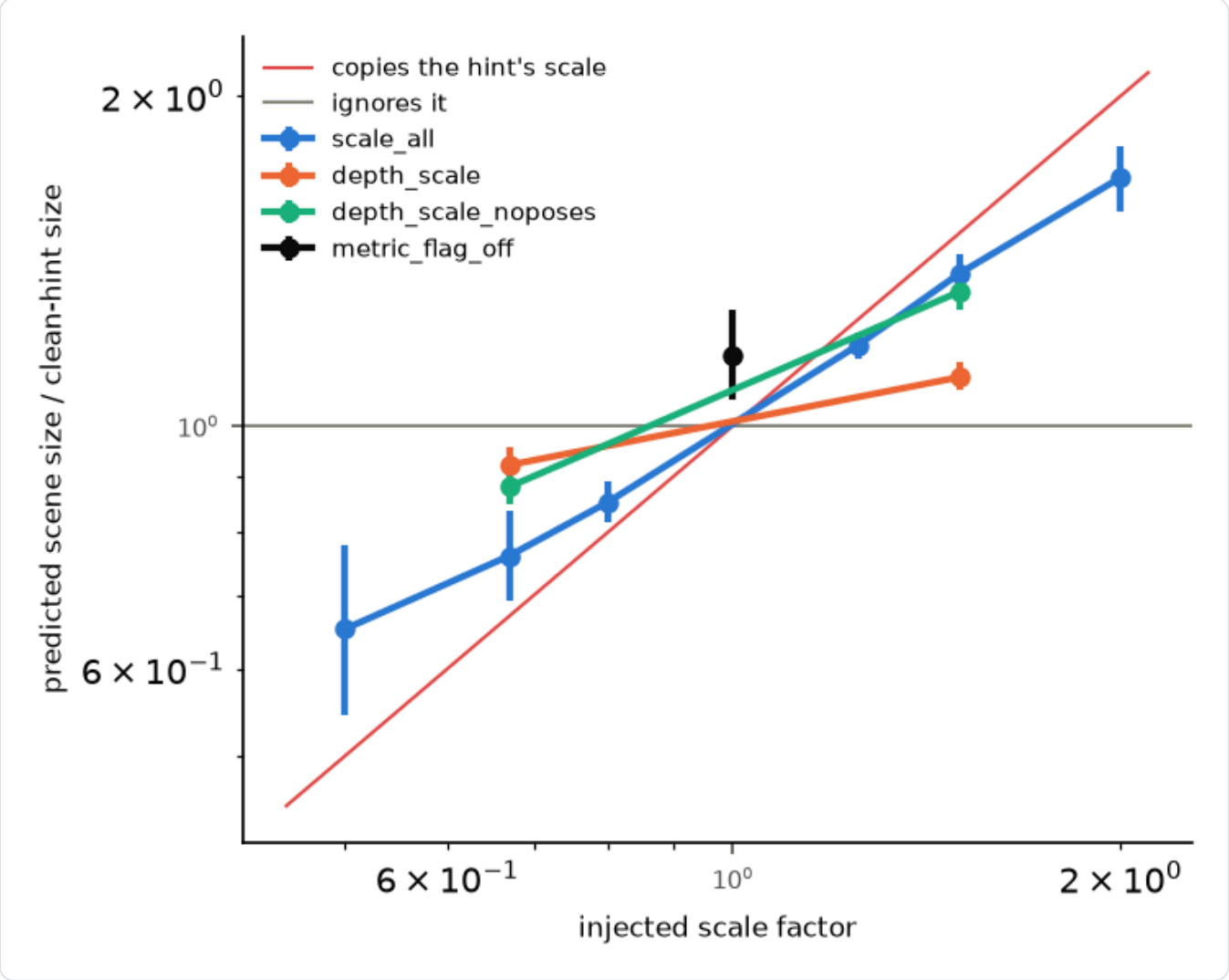
Error rise on the views whose hints were correct, compared with all-correct hints. Red line = the 20% bar for "spread".

Does the model copy bad hints, and does it notice?



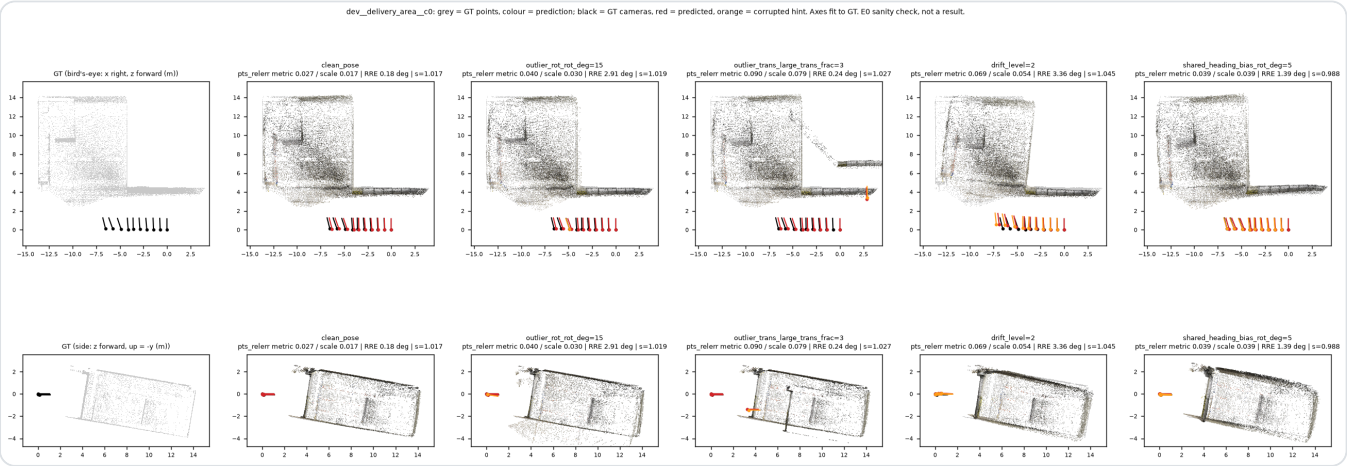
Absorption on the damaged view: 1 = the model copies the bad hint, 0 = it ignores it.

Wrong scale

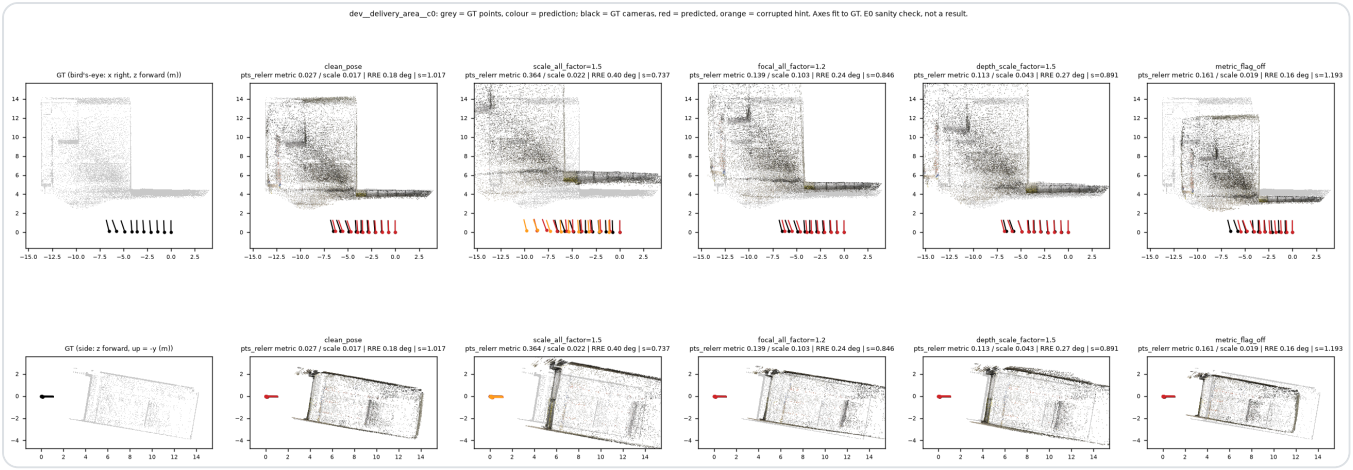


Predicted scene size relative to the correct-hint run, against the injected scale error. Red line = copies the wrong scale completely; grey line = ignores it.

What it looks like (practice scene, qualitative)



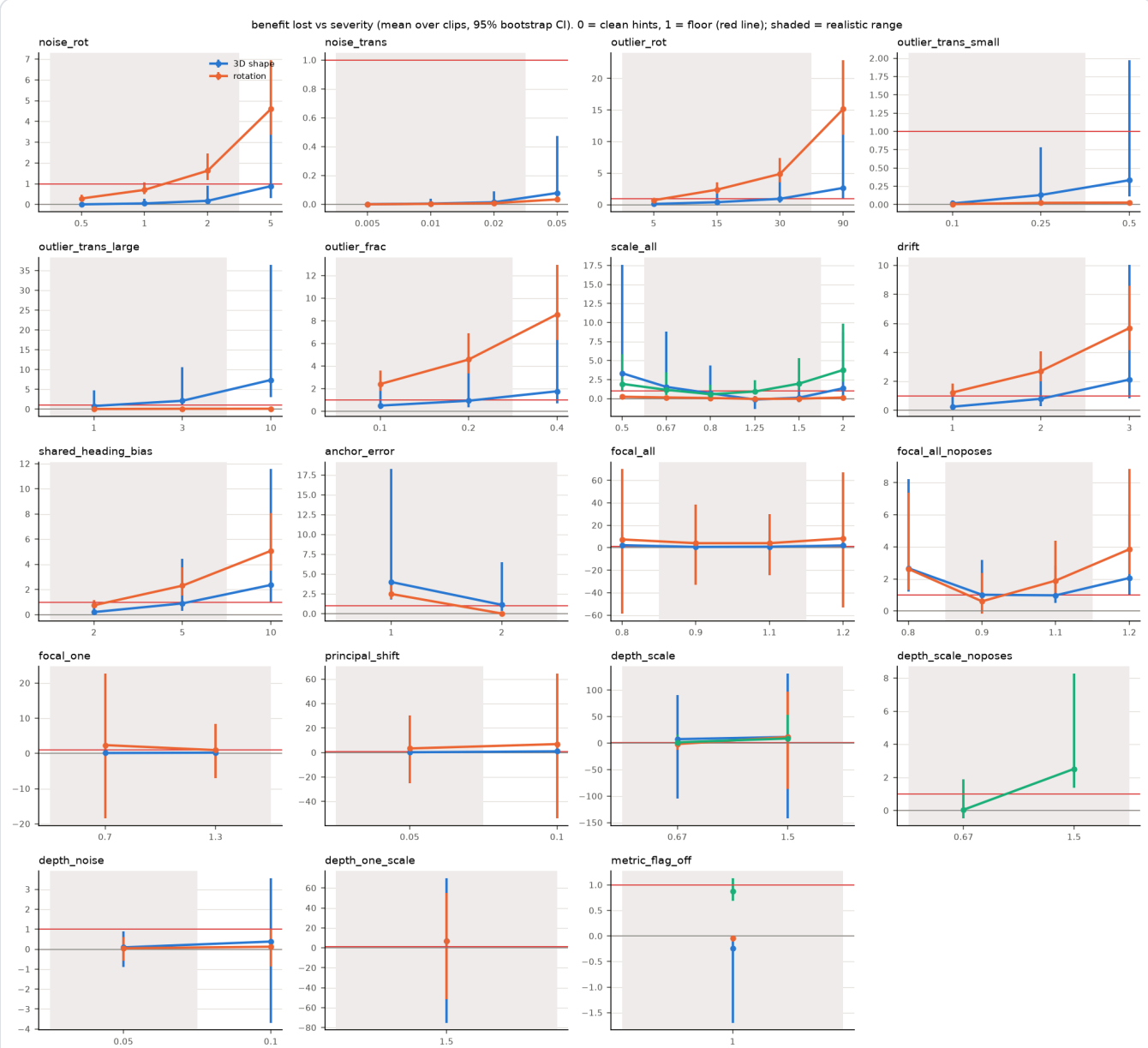
demo_e1_demo_pose.png: dev clip delivery_area, qualitative; grey = laser truth, colour = MapAnything, black = true cameras, red = predicted, orange = the wrong hint.



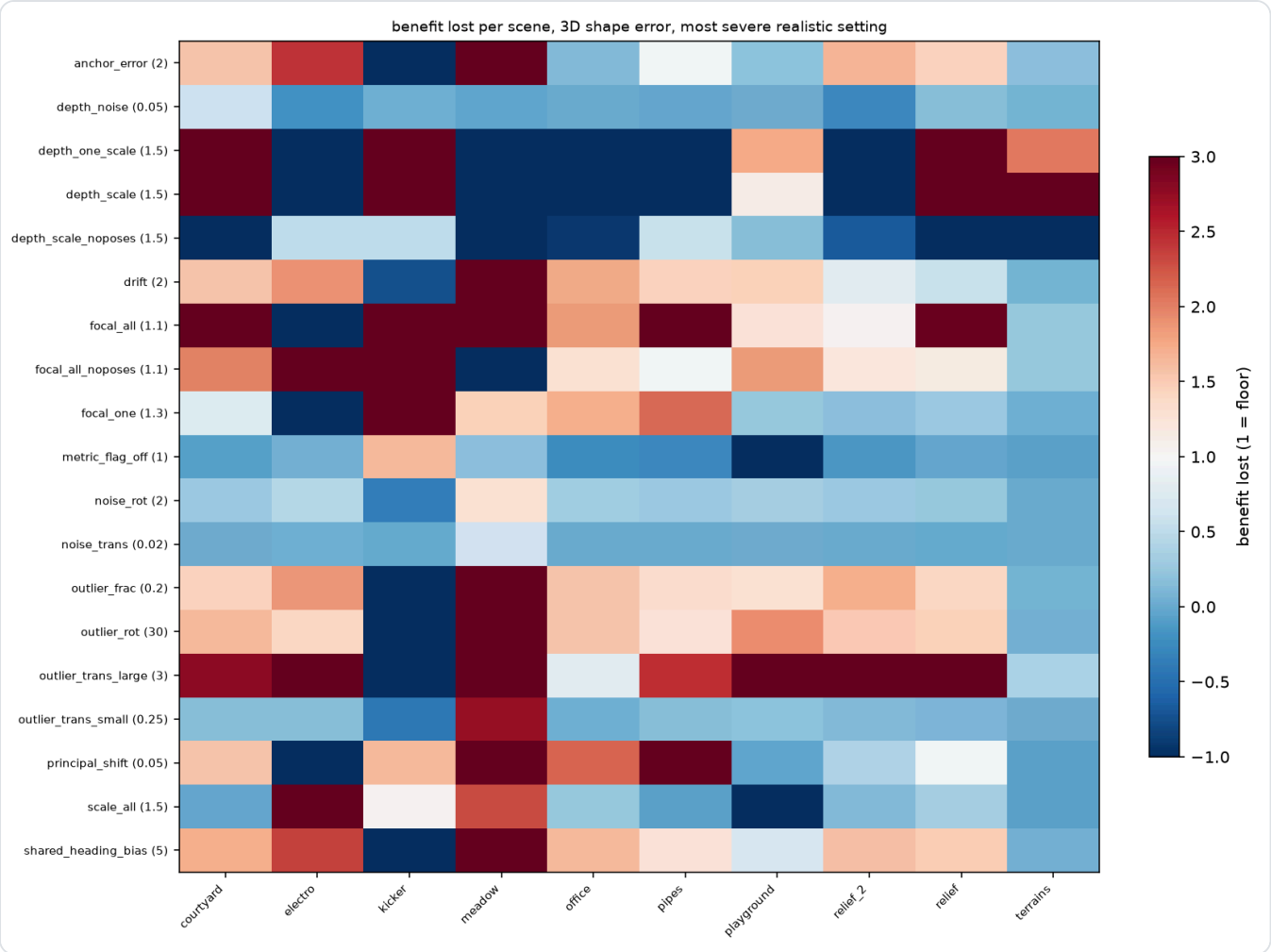
demo_e1_demo_other.png: dev clip delivery_area, qualitative; grey = laser truth, colour = MapAnything, black = true cameras, red = predicted, orange = the wrong hint.

As preregistered: "benefit lost"

The plan's own headline measure. It divides by how much the correct hint helped, so for lens and depth hints, where the correct hint helps very little, it swings wildly. Those panels have huge error bars, and the per-scene map shows random-looking stripes. The decision rules use it together with the 5% minimum effect; the figures above show the same comparisons in a stable form.



Benefit lost (0 = correct hint, 1 = no hint) against error size.



Benefit lost per scene (3D shape), most severe realistic setting.

Where to find everything

Technical report: `reports/e1_results.md`. Tables: `reports/e1_results/*.csv`. Raw runs: `results/e1_eval/`. Plan, frozen rules and amendments: `experiments/E1.md`.